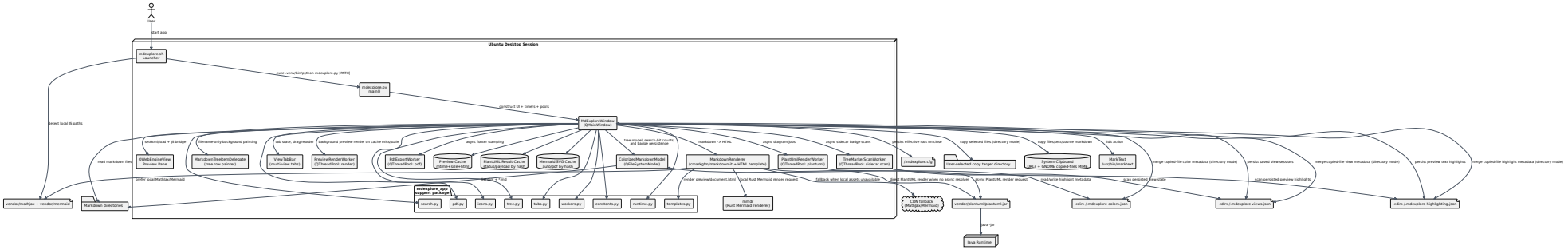
mdexplore UML

This document provides a subsystem-level PlantUML view of the current `mdexplore` implementation across `mdexplore.py`, `mdexplore.sh`, and `mdexplore_app/*`. All diagrams are embedded so they can be rendered directly by mdexplore (or any Markdown viewer with PlantUML support).

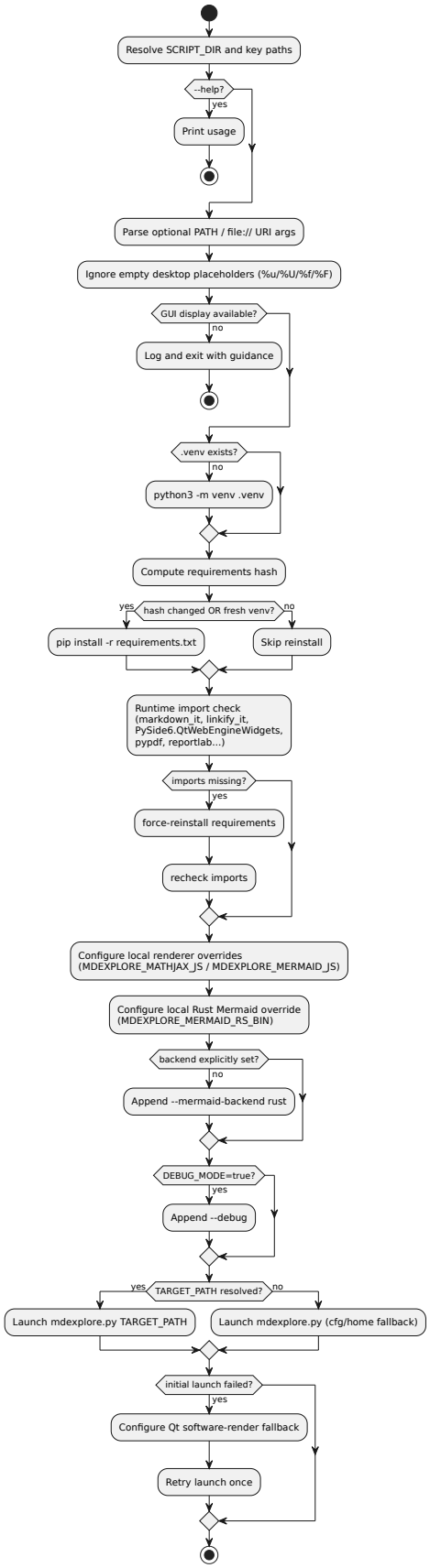
Detailed render/caching forks (GUI vs PDF, JS Mermaid vs Rust Mermaid, cache mode ownership, and restore behavior) are intentionally consolidated in the deeper render/debugging sections of `DEVELOPERS-AGENTS.md`. This UML file keeps those areas abstracted at system/class boundaries to avoid duplicating low-level render logic across two docs.

Low-risk modularization note: support code now lives in `mdexplore_app/` (`constants.py`, `runtime.py`, `search.py`, `templates.py`, `pdf.py`, `icons.py`, `workers.py`, `tree.py`, `tabs.py`), while the main orchestration, renderer/template wiring, and UI state machine remain in `mdexplore.py`. The diagrams below show that split at subsystem boundaries rather than expanding every extracted helper inline.

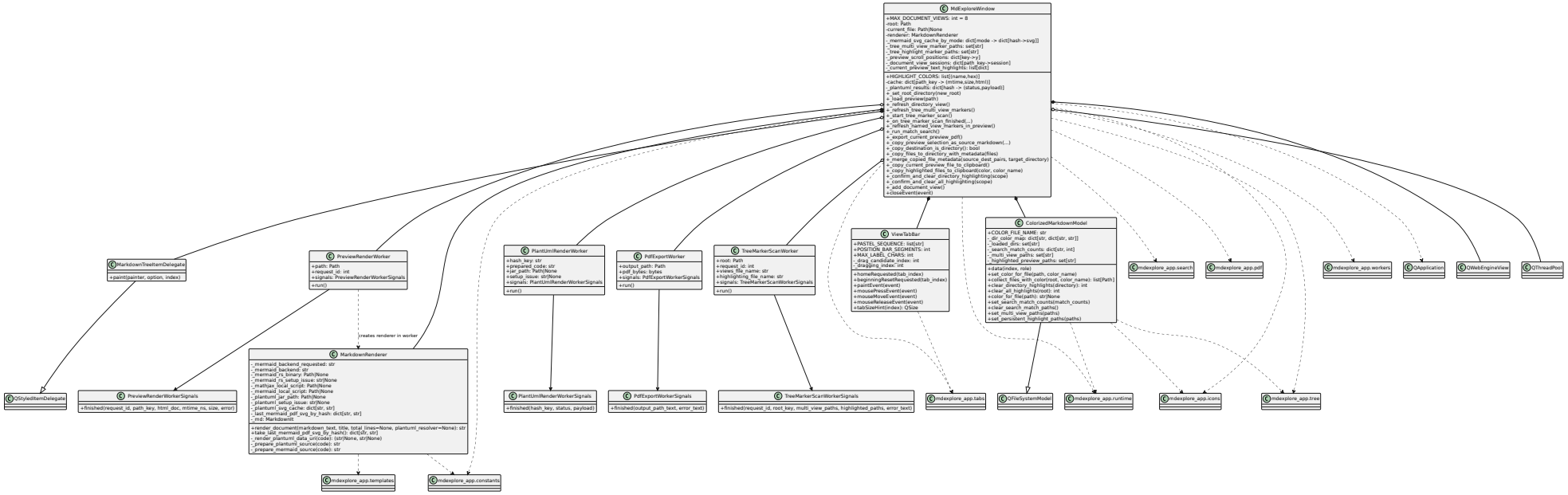
1. System Architecture



2. Launcher Runtime Activity (`mdexplore.sh`)

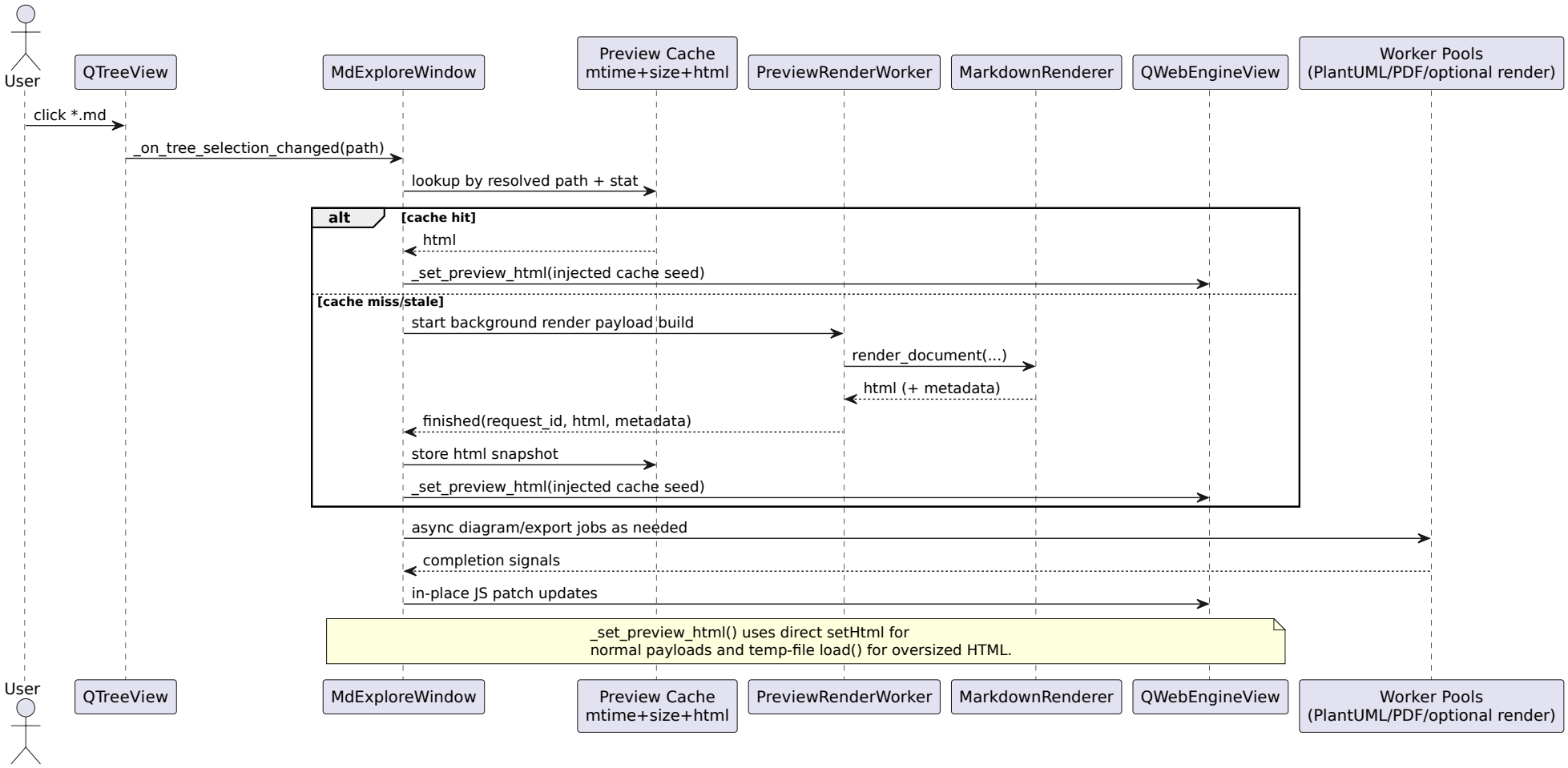


3. Core Class Diagram



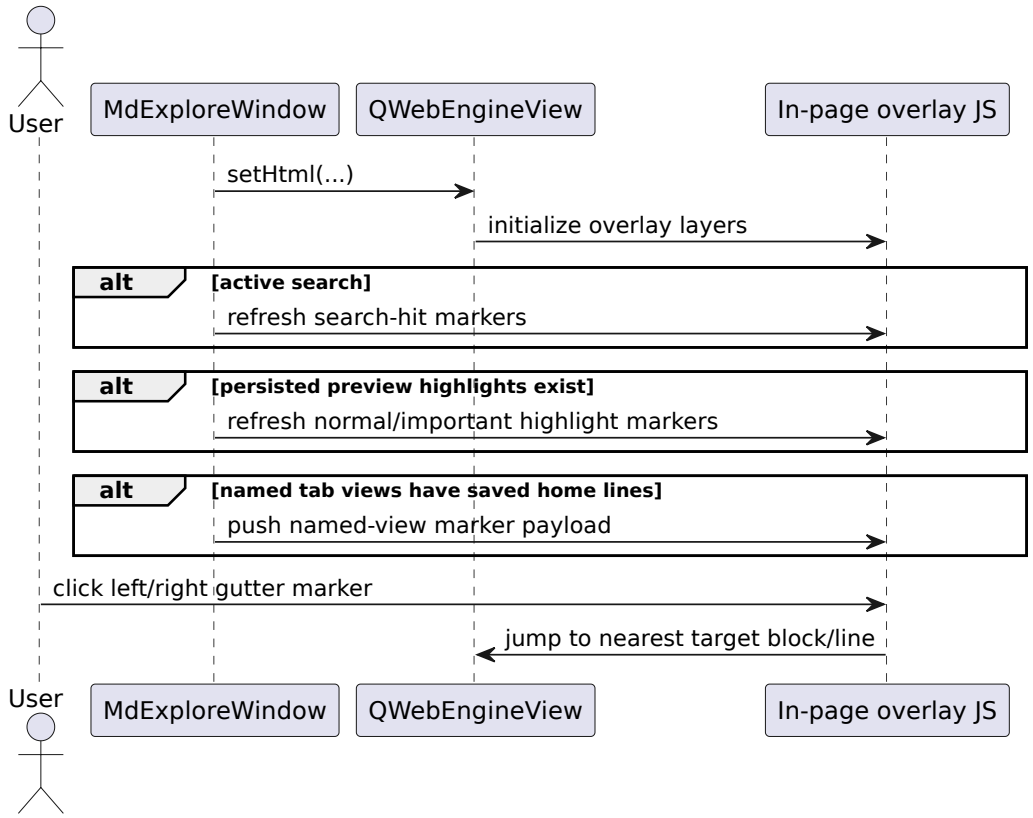
4. Preview Pipeline (Abstracted)

The detailed render/cache branches are covered in the render/debugging sections of `DEVELOPERS-AGENTS.md`. This section intentionally keeps an architectural boundary view only.

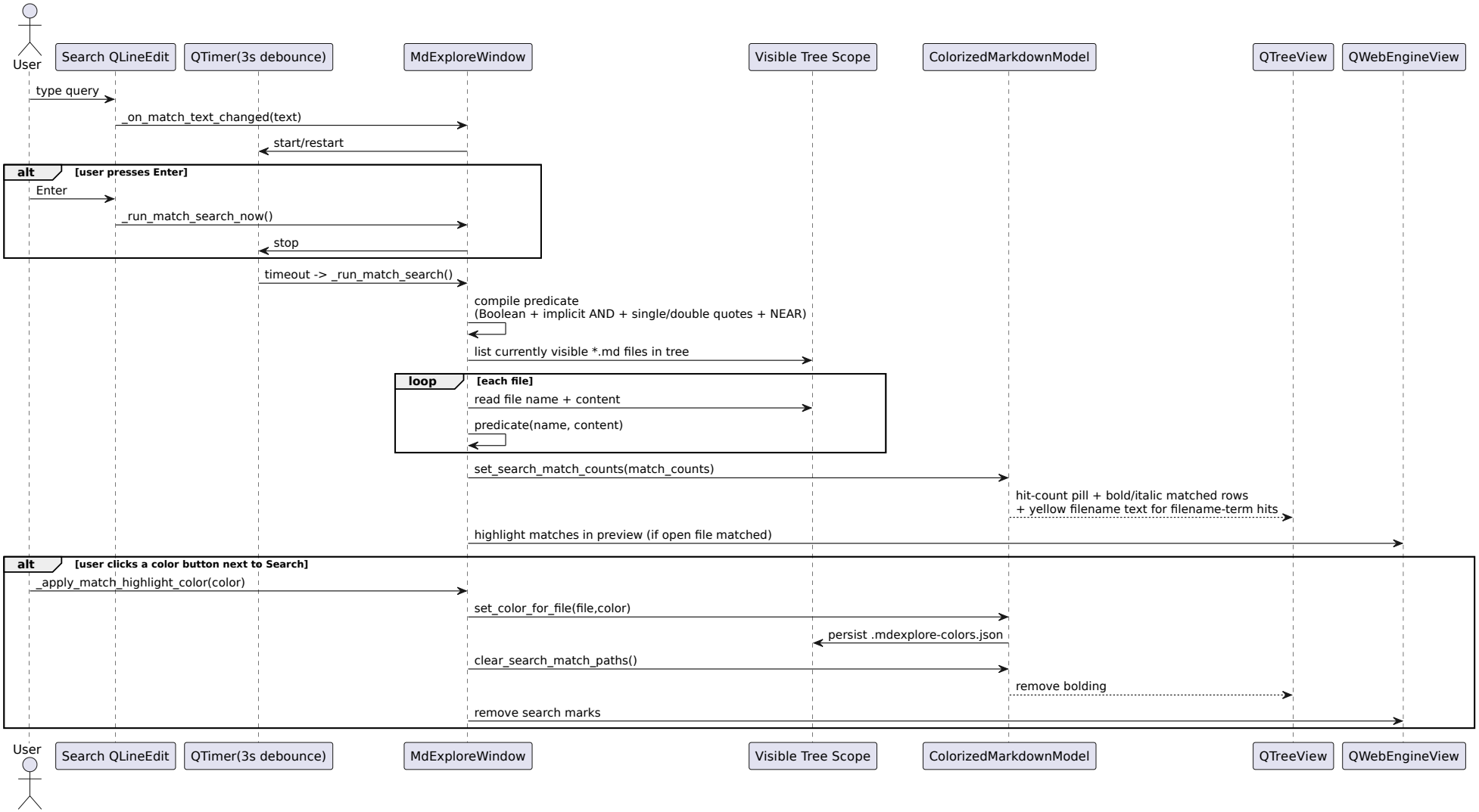


4.1 Preview Navigation Overlays

These are intentionally shown here only as boundary-level UI overlays. The exact DOM/CSS/JS mechanics remain in the deeper render/debugging sections of `DEVELOPERS-AGENTS.md`.



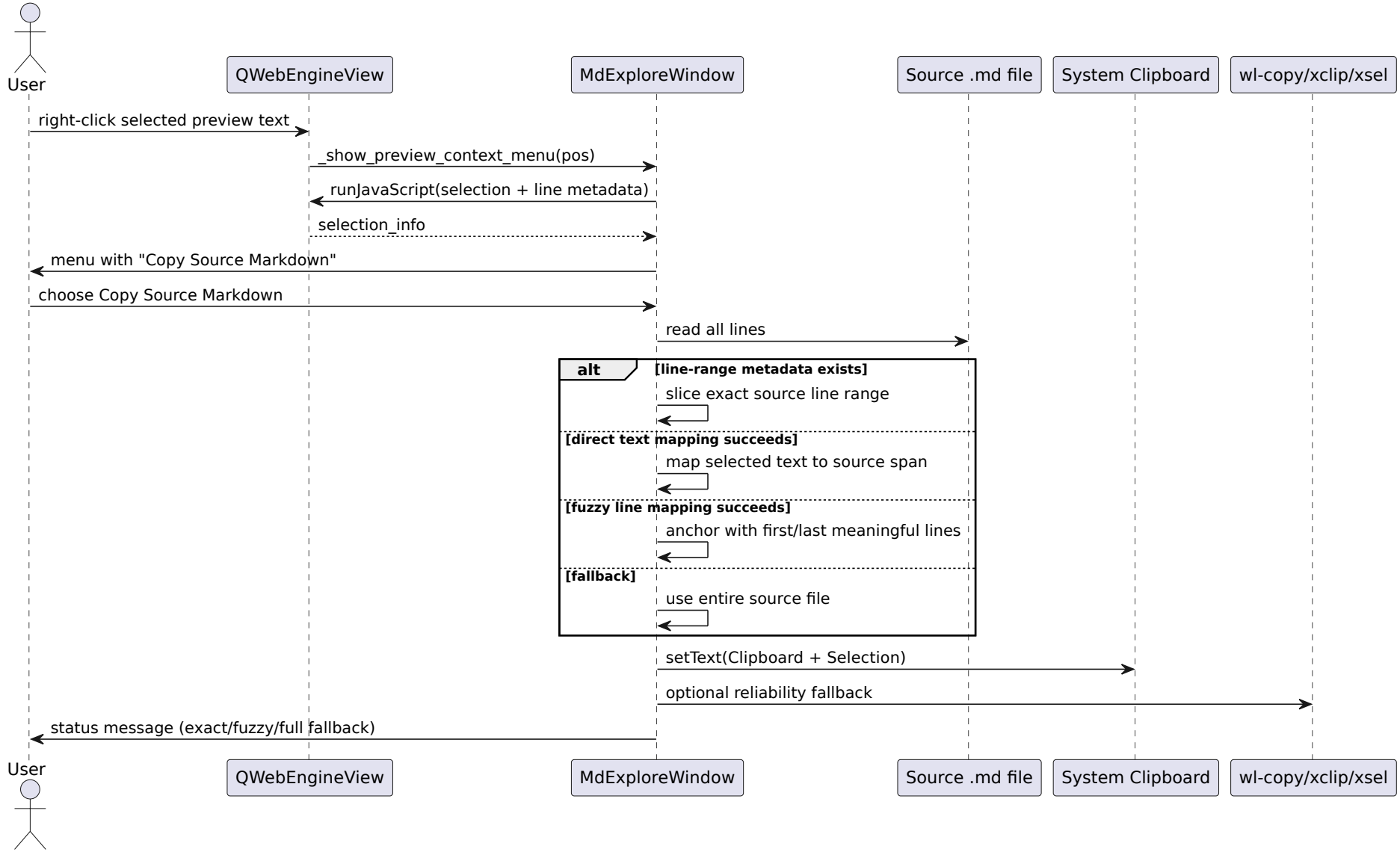
5. Search + Highlight + Apply Color Flow



Search semantics note:

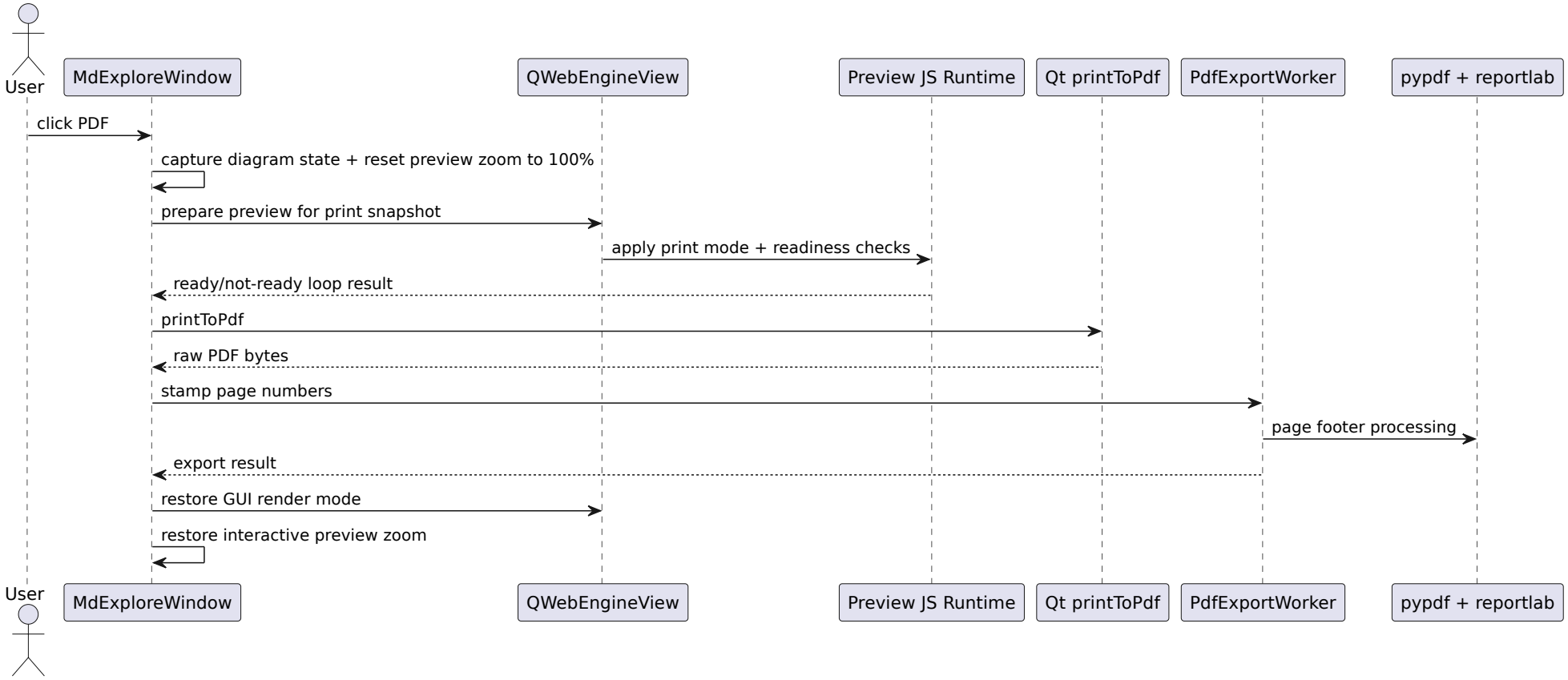
- unquoted terms are case-insensitive,
- double-quoted phrases are case-insensitive and preserve spaces,
- single-quoted phrases are case-sensitive and preserve spaces,
- only the opening quote character closes a quoted term, so apostrophes inside double-quoted phrases stay literal,
- `NEAR(...)` requires distinct qualifying occurrences per term, and single-word NEAR terms use word boundaries for proximity matching.

6. Preview Context Menu: Copy Source Markdown

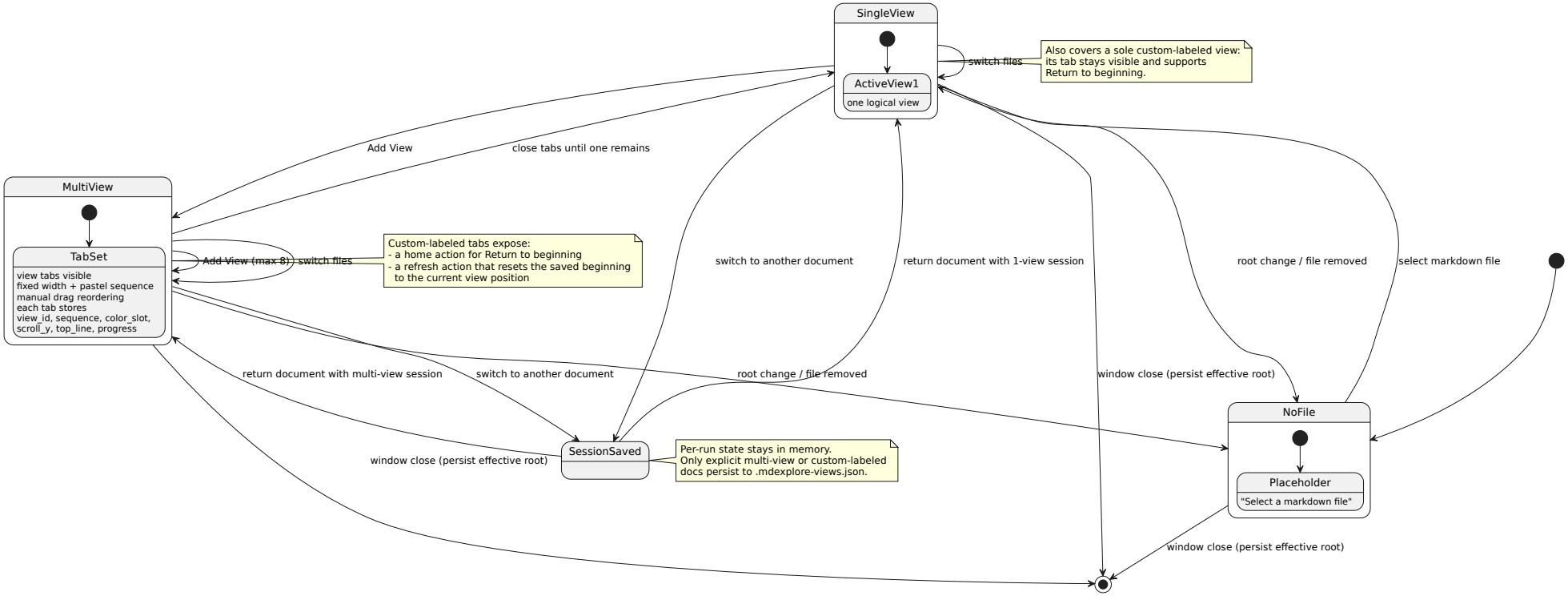


7. PDF Export Pipeline (Abstracted)

Detailed PDF mode branching (JS Mermaid grayscale path vs Rust Mermaid default-themed PDF cache path) is documented in the deeper render/debugging sections of `DEVELOPERS-AGENTS.md`.



8. Document View/Tab Lifecycle



Notes

- Diagrams are based on current code in `mdexplore.py`, `mdexplore.sh`, and `mdexplore_app/*`.
- Render/caching branch internals are intentionally abstracted here and documented in the deeper render/debugging sections of `DEVELOPERS-AGENTS.md` to keep a single authoritative deep map.
- Preview-only zoom (`Ctrl++`, `Ctrl+-`, `Ctrl+0`) is also intentionally kept out of the UML internals here because it is a `QWebEngineView` scale adjustment, not a separate renderer/cache branch.
- Preview gutter overlays (search-hit markers, persistent-highlight markers, named-view home markers) are similarly abstracted here and described in more detail in `DEVELOPERS-AGENTS.md`, because they are post-render navigation aids rather than renderer/cache forks.
- Search accepts canonical `NEAR(...)` syntax while continuing to treat `CLOSE(...)` as a backward-compatible alias normalized to `NEAR(...)` internally.
- Named-view gutter markers now route back through the same saved-view restore path as tab selection, so marker navigation and tab selection land on the same saved location.
- Effective-root directory styling is stateful in the tree:
  - bold aqua-blue when no active-search hits are under effective scope,
  - bold yellow with appended hit-count pill when active-search hits exist under effective scope.
- Top-right copy controls now include destination mode (`Clipboard` vs `Directory`); directory mode copies files into a chosen folder and merges copied-file metadata into destination `.mdexplore-*` sidecars.
- Headless regressions for saved-view restore, preview markers, and search quoting now live in `tests/test_preview_regressions.py` and `tests/test_search_query_syntax.py`, while template-asset regressions live in `tests/test_template_assets.py` and tab-bar layout regressions live in `tests/test_tab_bar_layout.py`.
- Worker/threadpool usage is intentionally separated by concern:
  - render pool (preview HTML jobs),
  - PlantUML pool (diagram jobs),
  - PDF pool (post-processing/stamping).
- PlantUML rendering is non-blocking in UI flow: placeholders are rendered first, then patched in place as jobs finish.
- TODO (known issue): diagram zoom/pan restore (Mermaid and PlantUML) is not yet consistently reliable when leaving a document and returning in the same app run.